Navaneeth Tanuboddi

AI Systems & Security Engineer • Full-Stack Developer

Bengaluru, India • navaneethtanuboddi@gmail.com • github.com/Navaneeth1324 • linkedin.com/in/navaneeth-tanuboddi

Education

PES University — Bengaluru, India

B.Tech in Computer Science & Engineering (AI & ML) • Researcher at PI Labs (Perception & Intelligence)

Relevant Coursework: Autonomous AI Agents, Machine Learning, Deep Learning, Data Structures & Algorithms, Cloud Security & RBAC, Operating Systems, Distributed Systems

Technical Skills

Languages:	Python, TypeScript, JavaScript, SQL, C++, Bash, HTML5/CSS3
AI & LLM Systems:	Parameter-Efficient Fine-Tuning (PEFT, QLoRA), PyTorch, Hugging Face, Agentic Tool-Use, Safety Evaluation (AgentHarm, ToolSafe)
Full-Stack & Web:	Next.js 14 (App Router), React 18, Tailwind CSS, Node.js, REST APIs, Context API, Recharts, Lucide Icons
Security & DevOps:	LLM Runtime Guardrails, Threat Modeling (OWASP LLM Top 10), Human-in-the-Loop (HITL), Git/GitHub, Linux

Featured Projects & Research

GuardAgent — Pre-Execution Guardrails for LLM Tools via MCP

github.com/Navaneeth1324/guard-agent

Tech: Python, Model Context Protocol (MCP), Mistral-7B (4-bit QLoRA), AST Parsing, Next.js 14 • *PES University Capstone (Guide: Ms. Ambika Kavital)*

- Architected a zero-trust pre-execution guardrail gateway utilizing **Model Context Protocol (MCP)**, intercepting 100% of LLM tool calls prior to OS/database socket dispatch to **eliminate prompt injections, RAG poisoning, and file wipeouts.**
- Engineered a two-tier hybrid inspection engine pairing sub-millisecond **AST syntax filtering (<0.1ms)** with a fine-tuned **4-bit NF4 Mistral-7B QLoRA model** (41.9M params), analyzing latent intent in **~138ms with a 100% fail-closed guarantee.**
- Benchmarked across **1,000 multi-turn test scenarios** (AgentHarm, ToolBench), achieving **95.2% Macro-F1** and **98.4% destructive recall**, significantly outperforming conversational baselines like **Llama Guard 3 (68.4% F1)** by eliminating tool blindness.
- Deployed an adaptive **tripartite policy router (Allow <0.35, HITL Review 0.35–0.70, Block ≥0.70)** across Cloud/DevOps (PocketOS volume wipeouts), Healthcare (HIPAA PII masking), and Banking, backed by a **real-time SSE telemetry monitor.**

PhonoShield — Real-Time Voice Deepfake & AI Audio Cloning Defense

github.com/Navaneeth1324/phonoshield

Tech: Python, Signal Processing, PyTorch, Librosa • *Acoustic Forensics Defense Framework*

- Developed an acoustic forensics framework detecting synthetic voice clones and neural vocoders via acoustic laryngeal micro-tremor analysis and higher-order spectral phase statistics.
- Extracted bispectral phase coherence cues and jitter/shimmer variances to differentiate biological vocal biomechanics from generative speech models (ElevenLabs, VALL-E) under sub-50ms streaming constraints.

GazeShield — AI Eye-Gaze Attention & Anti-Shoulder-Surfing Privacy Guard

github.com/Navaneeth1324/gazeshield

Tech: Python, OpenCV, Computer Vision • *On-Device Real-Time Privacy System*

- Engineered an on-device computer vision pipeline using sub-pixel iris localization and 3D head pose estimation to dynamically detect unauthorized observers and secondary lines of sight.
- Implemented automated, low-latency screen obfuscation and selective UI blurring triggered within **25ms** of shoulder-surfing detection, maintaining zero off-device telemetry.

Full-Stack Mobility & Real-Time Dispatch Platform (Uber Clone)

github.com/Navaneeth1324/uber-clone

Tech: Next.js 14, TypeScript, Tailwind CSS, Leaflet Maps • *Real-Time Web Application*

- Built a full-stack mobility platform featuring interactive geospatial routing, dynamic ride-booking simulation, simulated driver dispatch radar, and real-time partner dashboards.
- Implemented custom coordinate interpolation and reactive state pipelines using React Context and modular UI architecture.

Research & Publications

"Parameter-Efficient, Low-Latency Runtime GuardAgent for Tool-Using LLM Agents via Adaptive Human Oversight" — Summer Research 2026

Tanuboddi, N., K S, N., Mohin, Mirza, A. A. • PES University PI Labs Research